

# Euclid's Elements — Book XI

## Proposition 25

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

*If a parallelepipedal solid is cut by a plane parallel to the opposite planes, then the base is to the base as the solid is to the solid.*

## 1 Purpose and Scope

Proposition XI.25 is the point at which Book XI passes from the geometry of individual parallelepipeds to a genuine theory of magnitude and proportion. XI.24 established the geometric structure of a parallelepiped bounded by three pairs of parallel planes. XI.25 asks for substantially more: when such a solid is cut by a plane parallel to a pair of opposite faces, the ratio of the two corresponding base parallelograms is the same as the ratio of the two resulting solids.

The classical proof is Eudoxian. Euclid does not assign numerical areas or volumes. Instead he constructs arbitrary equal multiples of the two base magnitudes and the two solid magnitudes, proves that their three-way comparisons agree, and invokes Book V, Definition 5.

That proof architecture creates the main formal difficulty of XI.25. It is not enough to show that one pair of adjacent slabs is equal when their widths are equal. The reconstruction must build a coherent magnitude system in which

- equal bases and equal solids are genuine geometric equivalence classes;
- strict comparison means a genuine proper-part relation;
- arbitrary positive multiples are represented geometrically by repeated adjacent slabs;
- comparison of base multiples and solid multiples is synchronized; and
- the final appeal to Eudoxus' definition is not made true by definition.

The current production source therefore has two large layers. The first is the local geometry of adjacent parallelepipedal slabs. The second turns that geometry into quotient magnitudes, strict comparison, repeated multiples, and finally a faithful Book V proportion statement.

No coordinates, numerical area, or numerical volume are introduced.

## 2 Euclid's Statement

Euclid's proposition states that if a parallelepipedal solid is cut by a plane parallel to a pair of opposite faces, then the two base faces are in the same ratio as the two corresponding solids.

Schematically, if a solid is divided into adjacent parts

$$S_1 \quad \text{and} \quad S_2$$

by a parallel section, with corresponding base parallelograms

$$B_1 \quad \text{and} \quad B_2,$$

then the conclusion is

$$B_1 : B_2 = S_1 : S_2.$$

The assertion is not numerical. The equality of ratios is to be understood in the Eudoxian sense of Book V, Definition 5: for arbitrary positive multiples, the comparison pattern of the two base multiples must agree with the comparison pattern of the corresponding solid multiples.

## 3 Classical Configuration

Euclid begins with one parallelepiped cut by a plane parallel to two opposite faces. The cut produces two adjacent parallelepipeds sharing the section plane.

For the formal reconstruction it is convenient to represent the same local geometry by three parallel section planes

$$\pi_0 \parallel \pi_1 \parallel \pi_2$$

and two fixed pairs of side planes

$$\rho_0 \parallel \rho_1, \quad \sigma_0 \parallel \sigma_1.$$

Each section plane contains four vertices:

|       | $\pi$   | $\rho$   | $\sigma$     |
|-------|---------|----------|--------------|
| $A_i$ | $\pi_i$ | $\rho_0$ | $\sigma_1$   |
| $B_i$ | $\pi_i$ | $\rho_0$ | $\sigma_0$   |
| $C_i$ | $\pi_i$ | $\rho_1$ | $\sigma_0$   |
| $D_i$ | $\pi_i$ | $\rho_1$ | $\sigma_1$ . |

Thus the first slab lies between  $\pi_0$  and  $\pi_1$ , the second between  $\pi_1$  and  $\pi_2$ , and the whole lies between  $\pi_0$  and  $\pi_2$ .

The ordered version of the configuration also records

$$\begin{aligned} A_0 - A_1 - A_2, & \quad B_0 - B_1 - B_2, \\ C_0 - C_1 - C_2, & \quad D_0 - D_1 - D_2, \end{aligned}$$

so that the middle section is genuinely internal to the whole solid.

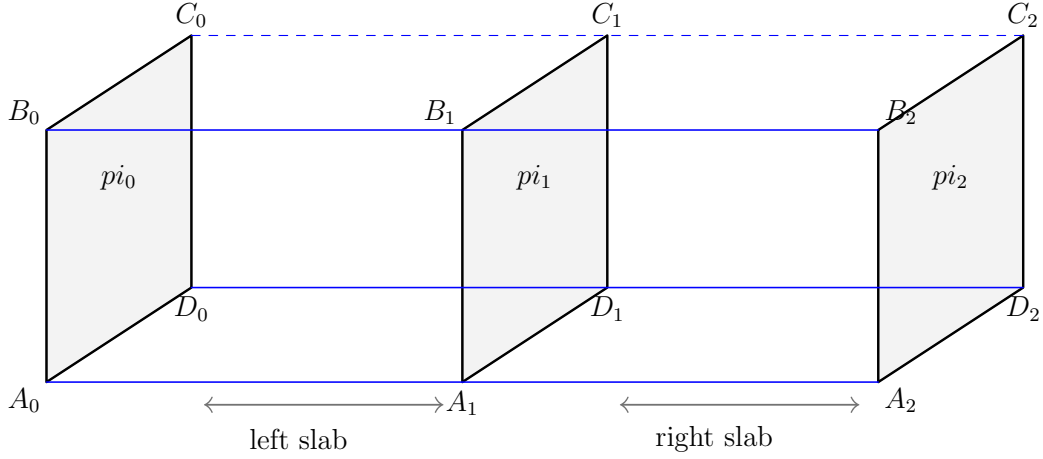


Figure 1: Three parallel sections determine two adjacent parallelepipedal slabs and their whole. Section edges are shown in black; the longitudinal spatial edges are shown in blue.

## 4 Euclid's Classical Proof

### 4.1 Equal repeated widths

Euclid extends one longitudinal direction on both sides of the original cut. He marks off any number of intervals equal to the width of the first slab and any number of intervals equal to the width of the second slab. Through the new division points he completes the corresponding parallelograms and parallelepipeds.

The purpose of the construction is not merely to obtain a larger solid. It constructs arbitrary positive multiples of four magnitudes simultaneously:

$$mB_1, \quad nB_2, \quad mS_1, \quad nS_2.$$

### 4.2 XI.24 supplies equal cells

For a chain of equal successive widths, XI.24 implies that the corresponding faces of successive elementary parallelepipeds are equal. Euclid then uses XI.Def.10 to conclude that the elementary solids are equal.

Thus a chain of  $m$  equal base cells corresponds to a chain of  $m$  equal solid cells, and similarly for the second family.

### 4.3 The comparison step

The decisive sentence in Euclid's proof is the three-way comparison:

$$\text{base multiple less than / equal to / greater than base multiple}$$

if and only if the corresponding solid multiple is less than / equal to / greater than the other solid multiple.

The equality case is supported by XI.Def.10. The strict cases use the geometric principle behind Common Notion 5: when one magnitude is represented as a genuine proper part of another,

the whole is greater than the part. Classically this proper-part geometry is largely left to the diagram.

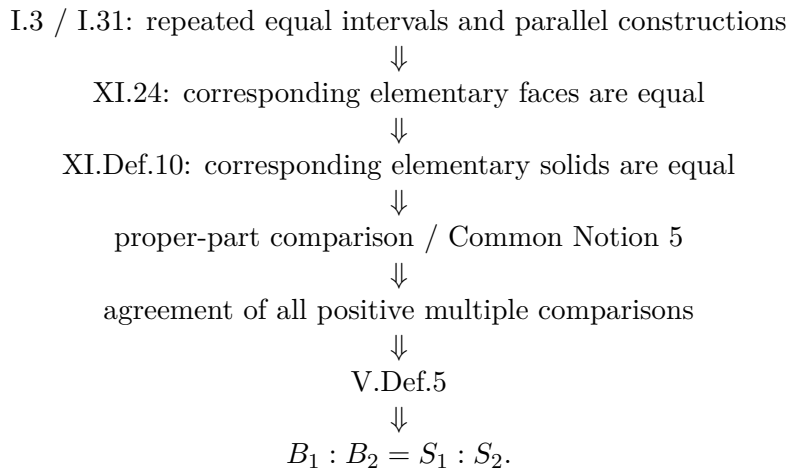
This is the point at which the formal reconstruction must do substantially more work than Euclid's text. It must produce an actual parallel cut that witnesses the smaller base and the smaller solid as proper parts of the larger ones; it cannot merely postulate a compatible order.

#### 4.4 Book V, Definition 5

Once comparison agreement has been established for arbitrary positive multiples, V.Def.5 gives

$$B_1 : B_2 = S_1 : S_2.$$

The classical dependency chain is therefore

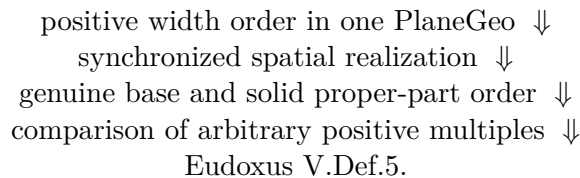


### 5 Why XI.25 Is Structurally Difficult

XI.25 is the first proposition in the present Book XI reconstruction whose main difficulty is not a single incidence or congruence argument. The proof must connect three different mathematical levels without identifying them: concrete three-dimensional parallelepipeds, geometric magnitude classes of bases and solids, and Eudoxian comparison of arbitrary positive multiples. The distinction is essential. A width parameter may organize the construction, but it must not define the very base and solid orders whose agreement XI.25 is supposed to prove.

This is also why the genuinely three-dimensional content cannot be hidden inside Book V. The proof needs three parallel section planes, two pairs of parallel side planes, four longitudinal edges, synchronized layoff of widths on those edges, construction of a section plane through three realized points, proof that the fourth point lies in that plane, and finally an actual cut exhibiting a smaller solid as a proper part of a larger one. The planar geometry **PlaneGeo** enters only where it should: widths are positive segment classes in one fixed carrier plane and their Book V order is handled there. These planar magnitude data are then realized back in ambient space.

The production development therefore has the following conceptual shape:



## 6 Formal Reconstruction: From One Cut to Eudoxus

### 6.1 The geometric core: two adjacent slabs and equal cells

The basic spatial object is

```
HilbertXI25TwoSlabConfiguration
```

with three parallel section planes, two pairs of parallel side planes, and twelve vertices. Its ordered refinement

```
HilbertXI25OrderedTwoSlabConfiguration
```

adds the four strict betweenness relations along the longitudinal edges. Thus the middle section is not merely another parallel plane: it is required to lie genuinely between the two outer sections on every longitudinal edge.

The direct route recognizes the two elementary slabs and their union through XI.24:

```
hilbert_XI25_twoSlab_XI24  
hilbert_XI25_whole_XI24
```

If two consecutive widths are equal on one longitudinal edge, the parallelogram-face information supplied by XI.24 propagates that equality to the other three longitudinal edges. This is recorded by

```
hilbert_XI25_equal_width_B  
hilbert_XI25_equal_width_C  
hilbert_XI25_equal_width_D  
hilbert_XI25_equal_width_all_edges
```

The corresponding adjacent side faces are then equal, and XI.Def.10 packages the resulting equality of elementary parallelepipeds:

```
hilbert_XI25_adjacent_rho_face_equal  
hilbert_XI25_adjacent_sigma_face_equal  
hilbert_XI25_adjacent_elementary_solids_equal
```

This is the formal version of Euclid's repeated use of equal successive intervals together with XI.24.

### 6.2 Magnitudes, proper parts, and positive multiples

The next step is not to attach numbers to bases or solids, but to form their geometric magnitude classes. A concrete parallelepiped is represented by

```
HilbertXI25Parallelepiped
```

and XI.Def.10 equality is encoded by

```
HilbertXI25SolidEquivalent  
HilbertXI25SolidClass
```

Bases are treated analogously by

```
HilbertXI25BaseEquivalent  
HilbertXI25BaseClass
```

so that equality of magnitudes is quotient equality rather than literal equality of records.

Strict comparison is geometric as well. The relations

```
HilbertXI25ProperBasePart  
HilbertXI25BaseLess  
HilbertXI25ProperSolidPart  
HilbertXI25SolidLess
```

are generated by an actual proper-part decomposition with a nonempty complement. The synchronized object

```
HilbertXI25AlignedProperPartWitness
```

records one cut that proves both the base comparison and the corresponding solid comparison. This is the precise formal content of the classical “whole greater than part” step.

Euclid’s positive multiples are then represented by actual chains of equal slabs rather than by an external volume function. The structures

```
HilbertXI25AlignedCutSum  
HilbertXI25RepeatedSlabMultiple  
HilbertXI25PositiveMultipleSystem  
HilbertXI25PairedPositiveMultipleSystems
```

keep base and solid multiples synchronized. For every pair of indices  $m, n$ , a comparison certificate records the common trichotomy: equality, left less than right, or right less than left, simultaneously for base classes and solid classes.

### 6.3 The noncircular comparison layer

This is the conceptual center of the reconstruction. A tempting formal shortcut would define both base order and solid order directly from the same width order. The final comparison agreement would then be true by construction and would no longer express Euclid’s theorem.

The production structure

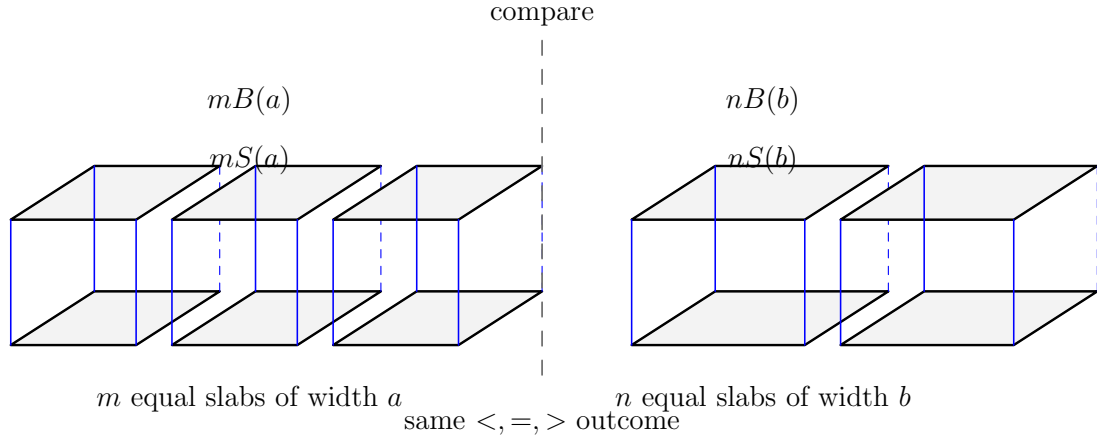


Figure 2: Positive multiples are represented by chains of adjacent equal-width slabs. The same geometric repetition generates the corresponding base and solid multiples.

`HilbertXI25ComparisonFaithfulFamily`

reverses that dependency. Widths retain the existing positive-segment order in one fixed `PlaneGeo`. Bases retain their geometric quotient order `HilbertXI25BaseClassLess`; solids retain `HilbertXI25SolidClassLess`. The family is accepted only after one proves that a strict width comparison produces genuine proper-part comparisons in both geometric systems. Trichotomy and asymmetry then show that base and solid comparison are faithfully reflected by the width source, not defined by it.

This distinction is what makes the final V.Def.5 argument substantive. The common width system supplies synchronized positive multiples, while the two target orders remain independently geometric.

## 6.4 Realizing arbitrary positive widths in space

A single nontrivial seed cut fixes the longitudinal direction and the four side planes. Every positive width class is first laid off on a fixed ray in `PlaneGeo`; Book Zero segment construction gives a canonical endpoint and the strict-less/betweenness theorem identifies the magnitude order with point order on that ray. The same width is then copied to the other three longitudinal rays by ambient segment construction.

For  $x < y$  the result is four synchronized betweenness relations

$$\begin{aligned} A_0 - A(x) - A(y), & \quad B_0 - B(x) - B(y), \\ C_0 - C(x) - C(y), & \quad D_0 - D(x) - D(y). \end{aligned}$$

The first three realized endpoints determine the section plane  $\pi(x)$ . The fourth point is not definitionally coplanar: its incidence with  $\pi(x)$  is proved using the spatial parallel-line machinery. XI.9 transfers the required line parallelism, XI.16 identifies the relevant section lines of parallel planes, and XI.15 proves that the resulting section plane is parallel to the original base plane. For  $x < y$ , XI.15 also yields  $\pi(x) \parallel \pi(y)$ .

The theorem

`orderedTwoSlabConfiguration_of_width_less`

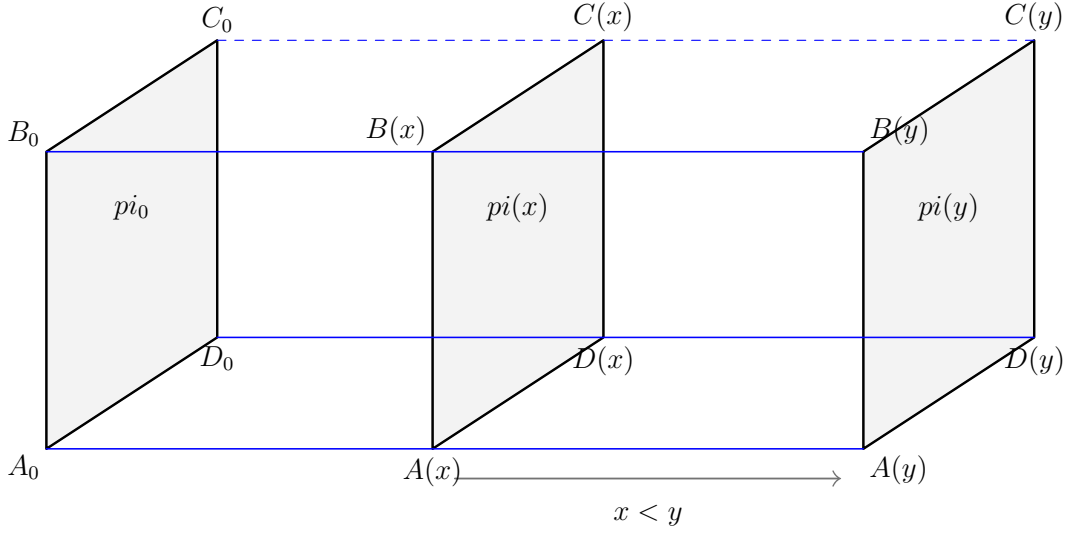


Figure 3: For  $x < y$ , one normalized width comparison in the carrier plane is transported to all four longitudinal edges. The realized endpoints determine two parallel section planes and an actual ordered cut.

turns these data into a complete ordered two-slab configuration, and

```
solidCutWitness_of_width_less
```

packages it as a genuine cut. The canonical direct-route representatives are built by

```
widthXI24
widthSolid
widthBase
```

and are proved equivalent to the left and whole pieces of the cut. Hence

```
alignedProperPartWitness_of_width_less
```

supplies exactly the geometric strict-comparison field required by the faithful family.

## 6.5 Closing the Eudoxus argument

The definition

```
HilbertXI25SolidCutWitness.toWidthRealization
```

packages the canonical base and solid representatives, the exact width specification, and the proper-part witness for every strict pair. From this, `toComparisonFaithfulFamily` provides the two geometric magnitude systems with their common positive-width source. The general theorem

```
eudoxusProportionBetween_of_common_source
```

then proves V.Def.5 without introducing numerical volume.

The terminal theorem of the direct production file

```
CGJteamLab/Proposition11_25.lean
```

is

```
Geometry.HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut
```

with conclusion

```
EudoxusProportionBetween
  ((X.toWidthRealization (Geo := Geo)).toComparisonFaithfulFamily
   (Geo := Geo)).baseEudoxusMagnitude
  ((X.toWidthRealization (Geo := Geo)).toComparisonFaithfulFamily
   (Geo := Geo)).solidEudoxusMagnitude
  a b a b
```

for arbitrary positive width classes  $a, b$  in the seed carrier plane. Thus one nontrivial cut generates not merely one ratio but a complete local family in which every positive pair satisfies the required base-to-base and solid-to-solid proportion.

## 7 The Hilbert–Wyler Route for XI.25

The separate production file

```
CGJteamLab/Wyler/Proposition11_25.lean
```

implements XI.25 through the Hilbert–Wyler route. It is important that this is not a second copy of the fifteen-thousand-line direct development. The neutral XI.25 structures, quotient magnitudes, proper-part relations, positive-multiple systems, faithful-family mechanism, section-plane construction, and final Book V argument are shared. The genuinely different part is the XI.24 front end and every XI.25 lemma whose proof depends on that front end.

The first replacement occurs at the level of the two adjacent slabs and their whole:

```
hilbert_XI25_twoSlab_XI24_wyler
hilbert_XI25_whole_XI24_wyler
```

These theorems use

```
euclid_proposition_11_24_wyler
```

instead of the direct XI.24 theorem. Consequently the six parallelogram faces are obtained by the Wyler XI.16 carrier-intersection argument developed for Proposition XI.24. The metric consequences remain neutral once the XI.24 face package has been produced.

The XI.25 geometry that actually consumes XI.24 is then rebuilt over this Wyler front end. Equal-width propagation is provided by

```
hilbert_XI25_equal_width_B_wyler
hilbert_XI25_equal_width_C_wyler
hilbert_XI25_equal_width_D_wyler
hilbert_XI25_equal_width_all_edges_wyler
```

and the equality of adjacent elementary solids by

```
hilbert_XI25_adjacent_elementary_solids_equal_wyler
hilbert_XI25_adjacent_solids_equal_of_base_equal_wyler
```

The quotient comparison layer is correspondingly connected by

```
hilbert_XI25_classComparisonCertificate_of_twoSlab_wyler
```

while the strict proper-part branches themselves remain the shared geometric ones.

The second essential replacement occurs in the canonical realization of an arbitrary width. The neutral construction of the points  $A(x)$ ,  $B(x)$ ,  $C(x)$ ,  $D(x)$ , their section plane, and all parallelism results is reused unchanged. What changes is the XI.24 package assigned to the resulting slab:

```
HilbertXI25SolidCutWitness.widthXI24_wyler
HilbertXI25SolidCutWitness.widthSolid_wyler
HilbertXI25SolidCutWitness.widthBase_wyler
```

The width specification is then re-established for these representatives by

```
HilbertXI25SolidCutWitness.widthBase_widthClass_wyler
```

and four equivalence lemmas identify the Wyler representatives with the left and whole pieces of the actual cut generated by  $x < y$ .

This yields the Wyler proper-part bridge

```
HilbertXI25SolidCutWitness.alignedProperPartWitness_of_width_less_wyler
```

and therefore the complete realization

```
HilbertXI25SolidCutWitness.toWidthRealization_wyler
```

At this point the two routes rejoin completely. The comparison-faithful family and the Eudoxus

theorem do not need a second Wyler-specific magnitude theory. They consume the same abstract `HilbertXI25WidthRealization` interface.

The terminal Wyler theorem is

```
Geometry.HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut_wyler
```

It has the same mathematical conclusion as the direct theorem, but its canonical solids and bases are generated through the Wyler XI.24 carrier route. This separation is architecturally significant: the Wyler method changes the spatial incidence proof where it genuinely differs, while Book V, quotient magnitudes, proper-part order, and the final Eudoxian logic remain shared.

## 8 Proof Architecture and Formal Obligations

The whole reconstruction is easier to read as one continuous argument than as a ladder of tiny formal layers. Starting from one nontrivial seed cut, the proof proceeds as follows.

1. XI.24 recognizes the two adjacent slabs and their whole and supplies the parallelogram-face data needed to transport equal consecutive widths.
2. Equal widths determine equal elementary faces and then equal elementary solids in the sense of XI.Def.10. Concrete bases and solids are passed to quotient magnitude classes, while strict comparison is defined by actual proper-part decompositions.
3. Repeated equal slabs realize the arbitrary positive multiples required by Book V. Their base and solid comparisons are synchronized by one geometric comparison certificate.
4. The comparison-faithful family prevents circularity: width order is a source order, whereas base and solid orders remain genuine geometric proper-part orders.
5. Every positive width class is laid off canonically on one carrier-plane ray and copied to the other three longitudinal rays. XI.9, XI.16 and XI.15 turn the synchronized endpoints into an actual parallel section.
6. For  $x < y$ , the two realized sections produce a genuine ordered cut. The smaller canonical base and solid are equivalent to the left piece, and the larger ones to the whole, so strict width comparison generates actual base and solid proper-part comparison.
7. The resulting `HilbertXI25WidthRealization` feeds the shared comparison-faithful family. Since all positive multiple comparisons agree, the common-source Eudoxus theorem yields V.Def.5.

Several facts that are visually implicit in Euclid's diagram must therefore be explicit formal obligations. The middle section must be internal on all four longitudinal edges; equal widths must be synchronized on four rays; the fourth realized endpoint must be proved coplanar with the first three; two section planes obtained from  $x < y$  must be proved distinct before their parallelism is used; a proper part must carry a nonempty complementary piece; and magnitude equality must be quotient equality rather than equality of chosen records. Finally, V.Def.5 quantifies over arbitrary positive multiples, so the proof cannot stop at one pair of adjacent slabs.

These are not implementation accidents. They are precisely the pieces of mathematical information compressed by the classical diagram and by Euclid's short comparison argument.

## 9 Dependencies and Reuse

The most visible dependency is XI.24: it supplies the complete face package for every elementary or canonical parallelepiped used by XI.25. In the direct route this is `euclid_proposition_11_24`; in the Wyler route it is `euclid_proposition_11_24_wyler`. Earlier Book XI results enter the arbitrary-width construction more locally: XI.9 transports parallelism of spatial lines, XI.16 identifies section lines of parallel planes, and XI.15 proves the realized section planes parallel to the base and to one another.

Book V supplies the magnitude-theoretic target: positive segment classes, positive multiples, trichotomy, comparison of multiples, and Eudoxus' Definition 5. XI.25 does not replace this by a numerical area or volume model; it proves that the geometric base and solid magnitude systems have the same comparison pattern.

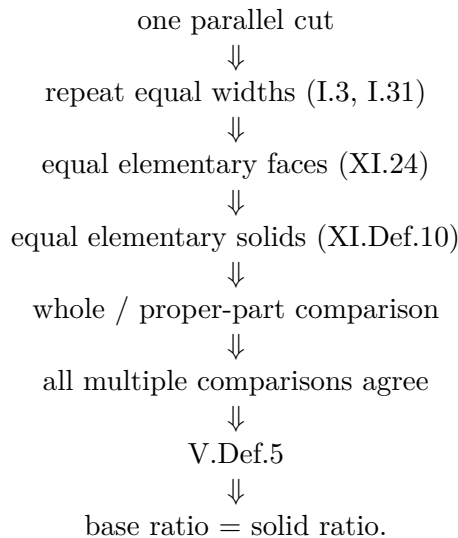
Book I contributes the planar parallelogram and congruence results inherited through XI.24. Book Zero supplies the canonical segment-layoff machinery and the strict-less/betweenness bridge used to realize arbitrary positive widths on a fixed ray. The ambient spatial proof uses the established

```
HilbertIncidence
HilbertOrder
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

interfaces, while planar magnitude reasoning is localized to explicit `PlaneGeo` slices.

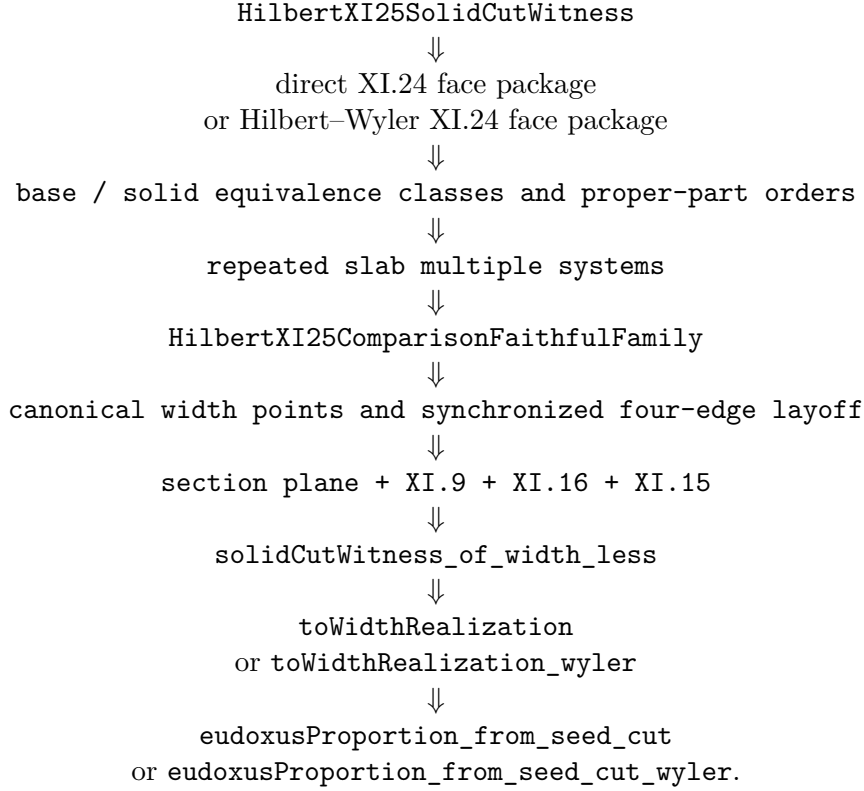
## 10 Classical and Formal Dependency DAGs

The classical chain is



The production Lean chain is longer because every implicit comparison must be exposed. The direct and Hilbert–Wyler routes differ only at the XI.24 front end and at the canonical

representatives built from that front end:



The additional formal length is not numerical overhead. It is the explicit geometry of the comparison statement that Euclid leaves largely implicit. The Wyler reconstruction confirms that this long comparison layer is not tied to one incidence proof of XI.24: once the alternative carrier-based face package has been supplied, the same magnitude and Eudoxus architecture can be reused.

## 11 Comparison with Euclid

The reconstruction preserves Euclid’s essential architecture unusually closely at the highest level:

1. construct repeated equal geometric cells;
2. identify equal elementary solids through XI.Def.10;
3. compare arbitrary multiples;
4. apply V.Def.5.

The major difference is the treatment of strict comparison. Euclid states that if one extended base exceeds another, then the corresponding extended solid exceeds the other. The formal proof cannot accept this as a diagrammatic fact. It must build a real section plane, exhibit the smaller solid as a proper part of the larger solid, and prove that this proper part is not equivalent to the whole.

A second difference is that the production theorem develops a complete local width family from one seed cut. This gives a reusable formal structure in which arbitrary positive widths, not only the original two pieces, have canonical base and solid representatives.

A third difference is organizational. Euclid moves directly from repeated segments to repeated bases and solids. Lean inserts quotient layers so that “equal base” and “equal solid” become well-defined magnitude equalities independent of representatives.

These changes strengthen the logical transparency of the argument without replacing its synthetic content by coordinates or measure.

## 12 Diagrammatic Audit

Three figures are retained because they show three genuinely different stable states.

1. Figure 1 shows the local geometric cut: two adjacent slabs and their whole.
2. Figure 2 shows the repeated-cell mechanism behind arbitrary positive multiples and V.Def.5.
3. Figure 3 shows the later canonical realization of arbitrary ordered widths  $x < y$  on all four longitudinal edges.

No separate figure is introduced for quotient construction, setoid transport, asymmetry lemmas, or the final Eudoxus wrapper. Those steps change the logical organization of the magnitudes but do not create a new stable geometric state.

## 13 Audit Status

This documentation has been checked against both current production sources:

```
CGJteamLab/Proposition11_25.lean
CGJteamLab/Wyler/Proposition11_25.lean
```

The direct terminal theorem is `HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut`; the Hilbert–Wyler terminal theorem is `HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut_wyler`. Both production modules compile cleanly in the completed XI.25 development.

The separate kernel audit of the direct theorem reports no new proposition-specific geometric axiom. It uses the established Book Zero null-segment infrastructure required by the positive-segment magnitude theory, together with the standard logical quotient and choice infrastructure. The Wyler route changes the spatial XI.24 derivation and the representatives built from it; it does not introduce a second magnitude or Eudoxus theory.

## 14 Conclusion

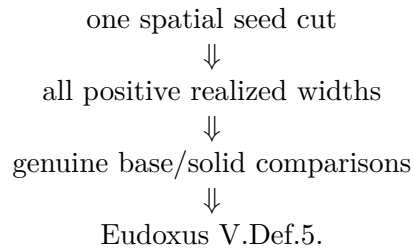
Proposition XI.25 is the first full volume-proportion theorem of Book XI and one of the most structurally revealing propositions in the reconstruction. It shows that Euclid’s proportion of bases and solids can be recovered without introducing numerical volume.

The proof proceeds instead through geometric equality, geometric proper parts, repeated equal slabs, and Eudoxus’ comparison of arbitrary multiples. The formal development makes explicit

the step that is most compressed in the classical text: strict comparison of bases and solids must be supported by a genuine cut and a nonempty remainder.

The decisive architectural achievement is the noncircular faithful family. Width classes organize the construction and provide a common multiple operation, but they do not define the target base and solid orders. Those orders remain geometric and are connected to width only by proved proper-part transfer.

Thus the final chain is genuinely synthetic:



XI.25 therefore marks the transition from the incidence and congruence theory of parallelepipeds to a formal synthetic theory of solid magnitude.